

Math 432 Implementation Homework 5

Due date: 24 May Sunday

You can get help from AI but write down your own code

1. (10 points) Write a code constructing the DDT of the S-boxes of DES.
2. (20 points) Construct a differential characteristic for 5-round DES on your own. The characteristic must be non-iterative and different from those presented in the literature. Write a program to experimentally measure the probability of your characteristic. The program should take the number of plaintext pairs as input and output both the number of pairs that produce the desired ciphertext difference and the corresponding empirical probability.